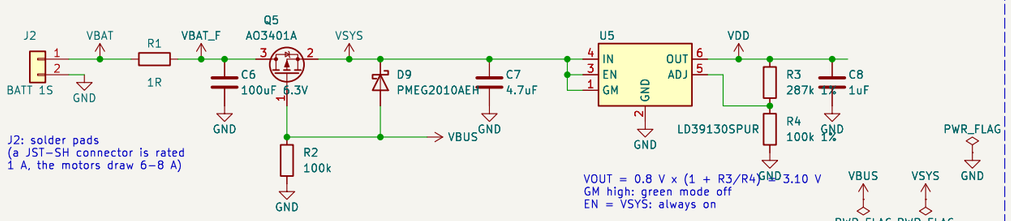
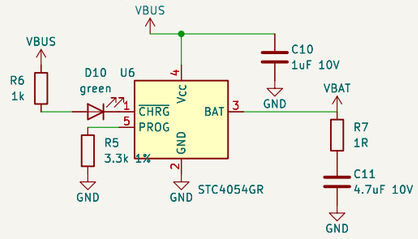
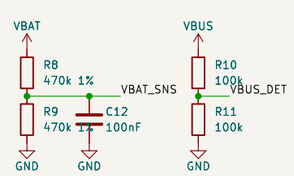
A — 1S battery, RC filter, USB/battery power path, LDO



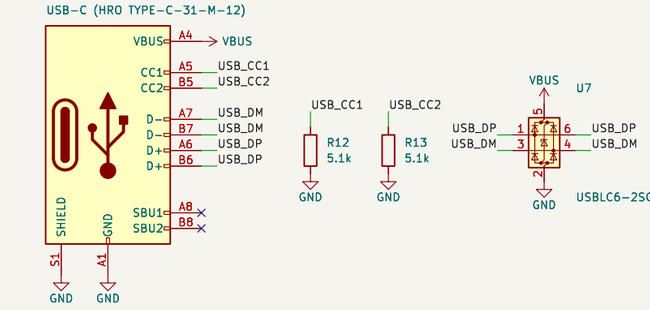
USB charger (STC4054, 4.2 V)



Sensing

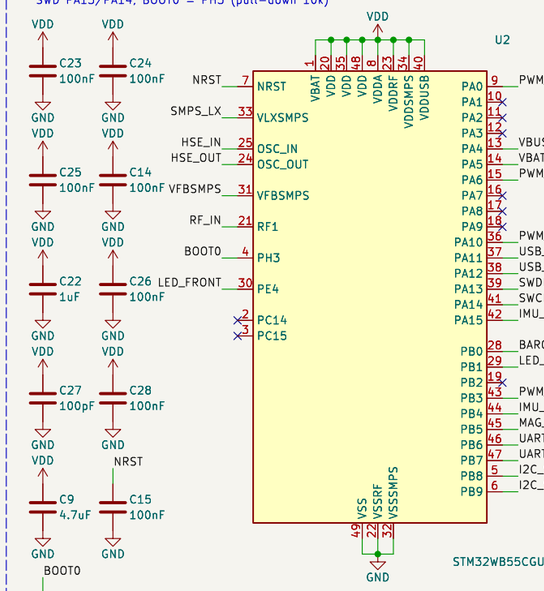


B — USB-C (charging + DFU/CDC), ESD protection

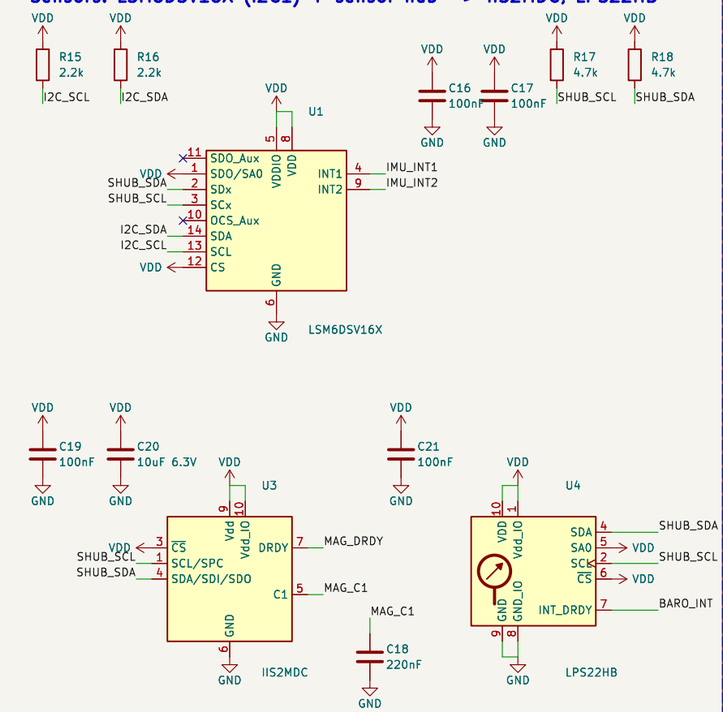


C — STM32WB55CGU6 (QFN48), RF front end, clocks

Pin map (IEKF firmware), rev C2:
Motor PWM: M1 PA0 TIM2_CH1, M2 PB3 TIM2_CH2,
M3 PA6 TIM16_CH1, M4 PA10 TIM1_CH3
LED: front PE4, rear PB1 (GPIO)
No LSE: RF wake-up HSE/1024, RTC HSE/32
ADC: PA5 VBAT_SNS (IN10), PA4 VBUS_DET (IN9)
I2C1 PB8/PB9 -> LSM6DSV16X (Mode 3)
IMU_INT1 PA15, IMU_INT2 PB4, MAG_DRDY PB5, BARO_INT PB0
USART1 PB6/PB7 -> TP1/TP2 (pogo), USB PA11/PA12
SWD PA13/PA14, BOOT0 = PH3 (pull-down 10k)

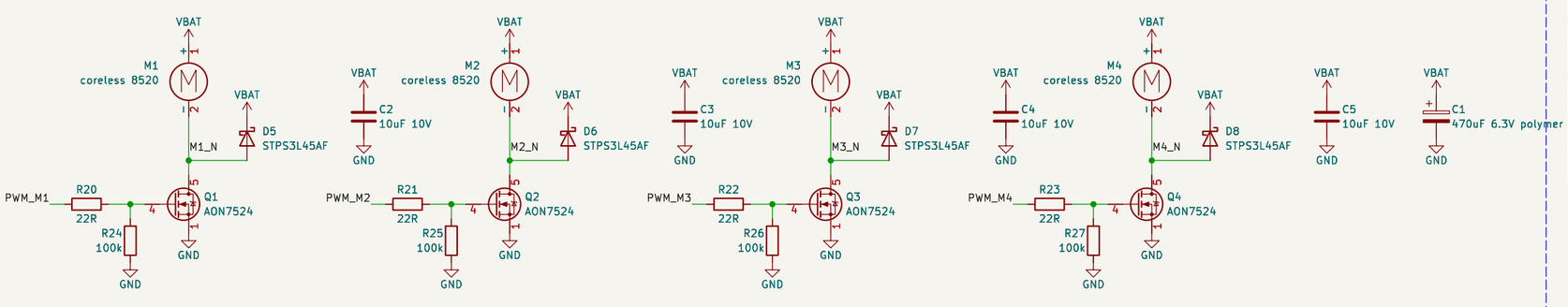


D — Sensors: LSM6DSV16X (I2C1) + sensor hub -> IIS2MDC, LPS22HB



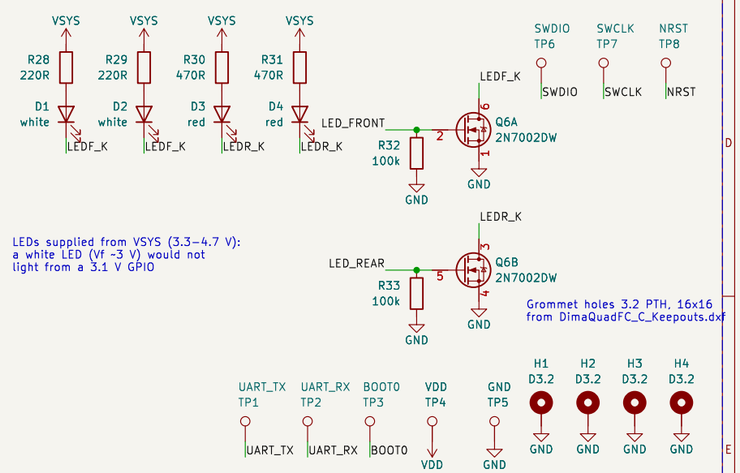
Topology = IKS4A1 Mode 3: the IEKF firmware
reads the IIS2MDC through the LSM6DSV16X
sensor hub (samples aligned in the FIFO).
Addresses: LSM6DSV16X 0x6B, IIS2MDC 0x1E, LPS22HB 0x5D

E — Motor stage: 4 x low-side N-MOSFET (BoM_motori.xlsx), 8520 coreless on 1S



Rg 22 ohm limits the GPIO peak current (Ciss 2250 pF):
Rpd 100k keeps the MOSFETs off during reset/boot.
PWM 32 kHz on TIM2 (ARR 2000 @ 64 MHz), same resolution as motors.c

F — Navigation LEDs, programming pads, mounting



Generated automatically: run ERC before release
Motor stage: BoM_motori.xlsx
Placement: DimaQuadFC_C_Placement.csv (mechanical handoff)
Outline and keep-outs: DimaQuadFC_C_Edge_Cuts.dxf, DimaQuadFC_C_Keepouts.dxf (frame C)

UAV Control — A. Cirillo

Sheet: /
File: DimaQuadFC_C2.kicad_sch

Title: DimaQuadFC rev C2 — 1S flight controller for frame C (STM32WB55CGU

Size: A3	Date: 2026-10-15	**Rev: C2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1